



(d) The distance between the slide and the screen is measured several times:

0.929 m 0.918 m 0.913 m 0.918 m 0.927 m.

Determine the mean distance D . Include the absolute uncertainty.

$D = \dots\dots\dots$ m [1]

(e) (i) Using your answers to (a), (c)(iii) and (d), determine the value of λ . Include an appropriate unit.

$\lambda = \dots\dots\dots$ [2]

(ii) Determine the percentage uncertainty in λ .

percentage uncertainty in $\lambda = \dots\dots\dots$ % [1]

(f) The experiment is repeated. Determine the slit separation s that gives a value of y of (0.500 ± 0.005) cm. Include the absolute uncertainty.

$s = \dots\dots\dots$ m [2]

[Total: 15]

